Chinatip Lawansuk（廖有福）

Taipei, Taiwan | Email: l.chinatip@gmail.com | Mobile: +886987422475 | Linkedin: Chinatip Lawansuk
Github: github.com/chinatip-l

Professional Summary

Driven by a passion for building robust systems from low level, I began my career as an Embedded System Engineer, where I developed firmware and infotainment systems for a fleet of over 1000 electric vehicles. To expand my knowledge from the system level to the chip level, I pursued a Master's degree in Electrical Engineering, focusing on digital IC design. This advanced study allowed me to dive deep into hardware design and optimisation, culminating in a thesis where I designed and implemented a CORDIC hardware accelerator with a custom caching system.

Expertises

Programming: Verilog, SystemVerilog, C, C++, Python, Assembly, Shell Script, TCL, Tensorflow
IC Design Software: Synopsys VCS, Synopsys Design Compiler (Design Vision), Synopsys TetraMAX, Cadence NC-Verilog, Cadence Innovus, Siemens Calibre
Skills: Cell-Based Digital IC Design Flow, Image Processing, Computer Architecture, Operating System, Embedded System Design, Firmware, RTOS, Circuit Design
Coursework: Computer-Aided VLSI System Design, Mixed-Signal IC Design, Computer Vision, Computer Architecture, Computer Organisation, Operating System

Research and Publication

CORDIC hardware Improvement using Content Addressable Memory Cache (Thesis)

- Investigated and designed an algorithm to cache partially calculated data and reuse it in the CORDIC architecture for sine and cosine calculation.
- Designed and implemented the processing core to suit the caching algorithm
- Designed and implemented a custom TCAM to use with the proposed cache unit and a complete cache subsystem from scratch, including multiple cache blocks, a PLRUm replacement policy, and a controller
- Designed and implemented a pipeline and synchronisation mechanism using a handshaking protocol
- Completed the full cell-based IC design flow from logic design and verification, synthesis, Design For Testability, Automatic Place and Route, post-layout simulation, and transistor-level simulation
- Using TSMC 90nm technology, resulting in a peak throughput of 67.796 MOPS, consumes 48.2 mW @ 338.98 MHz.

Two-Dimensional Convolution Accelerator of High-Precision Quantization (Conference Paper)

- Y. -T. Chao, J. -Y. Gao, **C. Lawansuk** and Y. -C. Fan, "Two-Dimensional Convolution Accelerator of High-Precision Quantization," *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA, 2024, pp. 1-2, doi: 10.1109/ICCE59016.2024.10444315.
- Developed a high-precision quantized 2D convolution accelerator leveraging the Winograd algorithm, achieving <0.2% quantization error.

Education

National Taipei University of Technology (國立臺北科技大學) — Sept 2023 –Aug 2025
MSc in Electrical Engineering (GPA: 4.00/4.00)
- **Laboratory:** Integrated System Laboratory (ISLAB) @ NTUT

King Mongkut's Institute of Technology Ladkrabang (KMITL) — Aug 2017 –Jul 2021
BE in Computer Engineering (GPA: 3.29/4.00)

Awards

First-Ranked Graduating Student (National Taipei University of Technology) — May 2025
Second Class Honour (KMITL) — Jul 2021

Experience

Firmware Engineer Intern, LinkCom Manufacturing Co., Ltd. – New Taipei, Taiwan — Aug 2024 - May 2025
- Worked on a high-power-density DC-DC Converter including hardware design (Planar Transformer, LLC topology) and embedded software optimisation. Performance reaches more than 1 kW on the credit card size module.
- Initiate the project for the transformer test platform (Firmware, and Hardware)
- Investigate and design closed loop control system for power module and identify impact parameters

Embedded System Engineer, MuvMi Co., Ltd. – Bangkok, Thailand — Oct 2021 - Sept 2023
- Develop IoT Fleet, Data Analysis and Visualisation Platform on AWS for EV Charging System, identified and mitigated up to 12% of energy loss
- Develop a Vehicle Control Unit Firmware (RTOS on arm Cortex M4 processor) and infotainment system (Buildroot on arm Cortex A7 processor) for Electric Minibus and Electric Tuk-Tuk for the fleet of 1000 vehicles

NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY
Taipei, Taiwan, R. O. C.

Name: Chinatip Lawansuk(Chinatip Lawa) Date of Birth: Feb. 16, 1999 Date Enrolled: Sep. 2023 Date Conferred: **** **** Date Issued: Feb. 20, 2025

Graduate Institute: International Program of Electrical Engineering and Computer Science(IEECS)

Degree Conferred: *** ***

The following transcript is hereby certified as correct according to the record of the university. Student ID: 112998405

Course	1st Semes. Credit	1st Semes. Grade	2nd Semes. Credit	2nd Semes. Grade
(2023 - 2024)				
Graduate Seminars	1.0#	94	1.0#	92
Detection and Estimation	3.0#	91		
Computer-Aided VLSI System Design and Practice	3.0#	92		
Operating Systems	3.0#	100		
Computer Vision	3.0#	90		
Deep Learning for Digital Image Analysis			3.0#	88
Error Control Technologies for Modern Mobile Communication Systems			3.0#	96
Mixed-Signal Integrated Circuit Design			3.0#	91
Data Science Principles with Applications on Educational Data			3.0#	87
Conduct	---	86	---	85
Total Credits & Average	13.0	93.3	13.0	90.6
(2024 - 2025)				
Master's Thesis	3.0#		3.0#	
Conduct	---	85	---	
Total Credits & Average	0.0	---		
Thesis				****
Total Credits "26.0"				
Total Average "91.96"				
GPA "4.00"				
Rank in Department : 1/8				
Total EMI Credits：26.0				
TO BE CONTINUED				

Remarks: The grading system is as follows: 80 or more=A; 70 to 79=B; 60 to 69=C; 50 to 59=D; less than 50=E; 70=the passing grade; W=Withdrawal; CW=Course Waived; P=Passing; F=Failing; #=English as a Medium of Instruction.

1912 NATIONAL TAIPEI UNIVERSITY OF TECHNOLOGY 教務專用章 國立臺北科技大學

Signature: Wan-Ting Huang
Registrar

Signature: Yuh-Shyan Hwang
Provost of Academic Affairs

B157238

KING MONGKUT'S INSTITUTE OF TECHNOLOGY LADKRABANG

Chalongkrung Road, Ladkrabang, Bangkok 10520, THAILAND

TRANSCRIPT OF RECORDS

Faculty of Engineering

Name : Mr. Chinatip LAWANSUK
Date of Birth : February 16, 1999
Program : Computer Engineering
Date of Admission : August 7, 2017

Student ID : 60010235
Degree : Bachelor of Engineering (Second Class Honors)
Major : -
Date of Graduation : June 14, 2021

Course	Credit	Grade
1st Semester, Academic Year 2017		
01006028 PRE-ACTIVITIES FOR ENGINEERS	1	A
01006030 CALCULUS 1	3	A
01076001 INTRODUCTION TO COMPUTER ENGINEERING	3	B+
01076002 PROGRAMMING FUNDAMENTAL	3	B+
90201001 FOUNDATION ENGLISH 1	3	A
90306003 LIVING SKILLS	3	A
GPS : 3.81 GPA : 3.81		
2nd Semester, Academic Year 2017		
01006031 CALCULUS 2	3	B
01076003 CIRCUITS AND ELECTRONICS	4	C+
01076004 OBJECT ORIENTED PROGRAMMING	3	B+
01076012 DISCRETE STRUCTURE	3	A
90201002 FOUNDATION ENGLISH 2	3	B+
90201030 ENGLISH FOR PROFESSIONAL PRESENTATION	3	B+
90401011 ENTREPRENEURSHIP	3	A
GPS : 3.38 GPA : 3.56		
1st Semester, Academic Year 2018		
01006032 ELEMENTARY DIFFERENTIAL EQUATIONS AND LINEAR ALGEBRA	3	B
01076005 DATA STRUCTURES AND ALGORITHM	3	C+
01076006 DIGITAL SYSTEM FUNDAMENTALS	4	B+
01076007 DATA COMMUNICATIONS	4	C+
90020008 ENGLISH FOR MARKETING	3	A
GPS : 3.08 GPA : 3.41		
2nd Semester, Academic Year 2018		
01076008 SOFTWARE DEVELOPMENT PROCESSES	4	C+
01076009 COMPUTER ORGANIZATION AND ASSEMBLY LANGUAGE	4	C+
01076010 COMPUTER NETWORKS	4	B+
01076253 PROBABILITY AND STATISTICS	3	B+
GPS : 2.96 GPA : 3.32		
1st Semester, Academic Year 2019		
01076011 OPERATING SYSTEMS	3	B+
01076021 COMPUTER ARCHITECTURE	4	C+
01076022 MICROCONTROLLER APPLICATION AND DEVELOPMENT	3	B+
01076026 USER EXPERIENCE AND USER INTERFACE DESIGN	3	C+
01076263 DATABASE SYSTEMS	3	C+
90593015 FUN WITH AI	3	B+
GPS : 2.97 GPA : 3.24		
2nd Semester, Academic Year 2019		
01076013 THEORY OF COMPUTATION	3	B
01076014 COMPUTER ENGINEERING PROJECT PREPARATION	1	A
01076023 COMPUTER HARDWARE DESIGN	3	B
01076025 SOFTWARE STUDIO	3	B
01076421 EMBEDED SYSTEM DESIGN	3	B+
01076533 DEEP LEARNING	3	B
01076582 ARTIFICIAL INTELLIGENCE	3	C+
90307001 THAI USAGE FOR COMMUNICATION	3	B
GPS : 3.04 GPA : 3.20		
Special Semester, Academic Year 2019		
01006004 INDUSTRIAL TRAINING	0	S
GPS : - GPA : 3.20		
1st Semester, Academic Year 2020		
01076015 COMPUTER ENGINEERING PROFESSIONAL DEVELOPMENT	2	A
01076311 PROJECT 1	3	B+
90401013 GENERAL BUSINESS	3	A
90591008 PHOTOGRAPHY APPRECIATION	3	C+
GPS : 3.45 GPA : 3.22		
2nd Semester, Academic Year 2020		
01076024 SOFTWARE ARCHITECTURE AND DESIGN	4	A
01076312 PROJECT 2	3	B+
90101002 MATHEMATICS IN DAILY LIFE	3	A
90108007 ENVIRONMENTAL STUDY	3	A
GPS : 3.88 GPA : 3.29		
Total Credits Earned : 135		
Cumulative GPA : 3.29		
------------------------------ End of Transcript ------------------------------		

Date of Issued : November 7, 2022

Not valid without seal

Pakorn W.

(Asst.Prof. Dr. Pakorn Watanachaturaporn)
Acting Director
Office of the Registrar

BA 0118988

ETS TOEIC

LISTENING AND READING TEST
OFFICIAL INSTITUTIONAL SCORE REPORT

Name CHINATIP LAWANSUK

Date of Birth FEBRUARY 16, 1999

ID Number 1100501503550

Test Date NOVEMBER 7, 2022

Client PERSONAL

LISTENING	TOTAL SCORE
460	875
READING	
415	

Report is valid for two years from the test administration date.

TOEIC® Services Thailand -- Certified Score Report -- Issued NOVEMBER 8, 2022

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TAIPEI TECH 國立臺北科技大學 National Taipei University of Technology

國立臺北科技大學獎狀

北科大教字第 1140100181 號

學生 Chinatip　Lawa 於 113 學年度 電資學院外國學生專班應屆畢業學業成績 第一名 特頒獎狀以資鼓勵

校　長 王錫福

中華民國 114 年 6 月 7 日

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